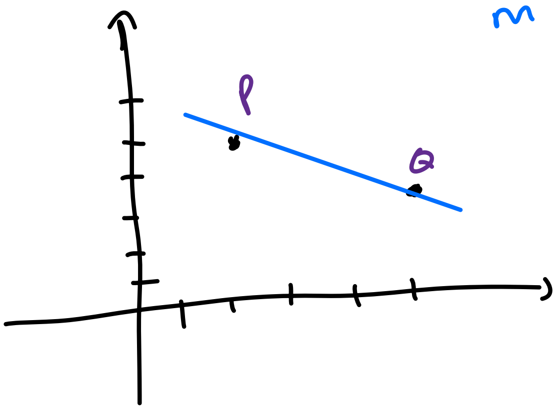
Agenda

1. Announcements
    - Regrade requests will be responded to by the end of the day on Monday (grades re-synced to Canvas on Tuesday)
    - Check for final exam conflicts – sign up for the makeup exam if needed
    - Please fill out course evaluations
2. Start Section 12.5

Section 12.5 - Equations of Lines and Planes

Parametric Equations of Lines:

Activity: Find the equation of the line L through the points P(2, 5) and Q(6, 3).



$m = -1/2$

$y - 5 = -1/2\,(x - 2)$

Example: Write down a vector equation of the line L through P(2, 5) and Q(6, 3).

need a point and a direction vector

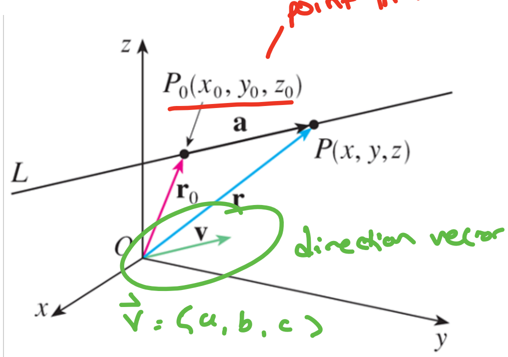
Point: $(2, 5)$

dir. vector: $\overrightarrow{PQ} = \langle 6-2,\ 3-5 \rangle = \langle 4, -2 \rangle = \vec{v}$

vector eqtn. of a line: $\vec{r}(t) = \underbrace{\begin{pmatrix} 2 \\ 5 \end{pmatrix}}_{\text{point}} + t\underbrace{\begin{pmatrix} 4 \\ -2 \end{pmatrix}}_{\text{direction vector}}$

output is vectors

More Generally:



$\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$

$\vec{r} = r_0 + \vec{a}$

$\vec{r}(t) = \vec{r}_0 + t\vec{v}$

ex
$\vec{r} = \langle 2, 5 \rangle$

in other words

vector eqtn. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underbrace{\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}}_{\text{point}} + t \underbrace{\begin{pmatrix} a \\ b \\ c \end{pmatrix}}_{\text{direction vector of line}}$

also written as

$$\begin{cases} x = x_0 + ta \\ y = y_0 + tb \\ z = z_0 + tc \end{cases}$$

parametric eqtn. of a line

or

$(t = t = t)$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Symmetric eqtn of a line.

Activity: Find the vector equation of the line through the point $(4, 3, -2)$ and parallel to the line

$x = 1 + 4t, \quad y = 2 + 5t, \quad z = -1 - t.$

$x = 1 + 4t$
$y = 2 + 5t$
$z = -1 - t$

rewrite parametric to vector eqtn

$\Downarrow$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix}$$

shares direction vector

want line thru $(4, 3, -2)$ with same direction

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \boxed{\begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}} + t \begin{pmatrix} 4 \\ 5 \\ -1 \end{pmatrix}$$

Geometry of Lines

In $\mathbb{R}^3$, two distinct lines can have three possible scenarios:

1.) parallel

2.) intersect at one point

3.) skew

(not parallel and never intersect)

not possible in $\mathbb{R}^2$

Example: Determine whether the lines $\vec{r}(t) = \langle 1+t, 1-t, 2t \rangle$ and $\vec{p}(t) = \langle 2-t, t, 2 \rangle$ intersect, are parallel, or are skew.

① Check if parallel

dir vector

$$r(t) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \boxed{\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}}$$

dir vector $p(t)$

$$p(t) = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + t \boxed{\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}}$$

cannot be multiples of eachother, and are not parallel.

② Intersect?

☆ NOTE: use different parameters for each line

$$\begin{array}{lccc} r(t) : & \langle \underbrace{1+t}_{x_1}, & \underbrace{1-t}_{y_1}, & \underbrace{2t}_{z_1} \rangle \\ p(s) : & \langle \underbrace{2-s}_{x_2}, & \underbrace{s}_{y_2}, & \underbrace{2}_{z_2} \rangle \end{array}$$

need:
$$\left.\begin{array}{l} 1+t = 2-s \\ 1-t = s \\ 2t = 2 \end{array}\right\} \text{all true}$$

③ $t = 1$

② $s = 0$

① 2 = 2 ✓✓ intersect

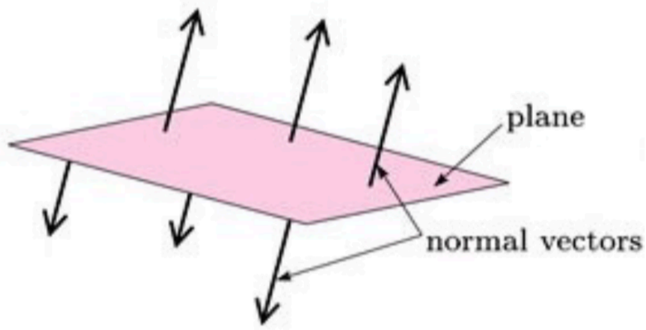
Intersect at (2, 0, 2)

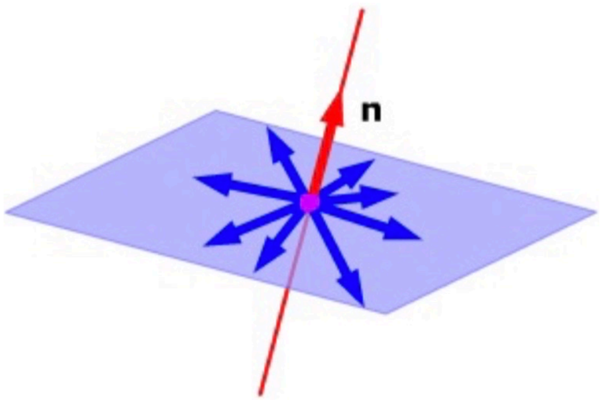
Notice: $\vec{r}(1) = \langle 2, 0, 2\rangle$

$\vec{p}(0) = \langle 2, 0, 2\rangle$

Planes

All vectors in a plane are orthogonal to the same direction, called the normal vector to the plane.





Activity: Find the equation of the plane that has normal vector $\langle -6, 5, 7 \rangle$ through the point $(1, 2, 3)$.

Example: Sketch the plane $2x + 3y + 6z = 6$ in the first octant of $\mathbb{R}^3$.

Three non-collinear points define exactly one plane in $\mathbb{R}^3$

<u>Example</u>: Find the equation of the plane through the points $P(0,1,2)$, $Q(-1,3,0)$, and $R(5,1,2)$.